

**The Birth of Morphotopy:  
From Earth-Measure to the Spaces of Life and Mind**

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## Abstract

In physics, biology, cognitive science, and machine learning, a recurrent procedure renders a non-spatial domain as a structured space of possible forms and treats its behaviour as constrained movement within that space; configuration spaces, morphospaces, conceptual spaces, and the latent spaces of trained networks are instances. We argue that this recurrence is the latest phase of a long process in the history of geometry, the progressive detachment of form from physical extension, and not a collection of independent metaphors. The history is traced stratigraphically, from land survey and the Euclidean figure through coordinates, non-Euclidean geometry, Riemann's intrinsic manifold, Klein's transformation groups, topology, phase space, and information geometry, to biological morphospace and the representational geometry of brains and machines, with each stratum remaining in use beneath the next. We name the current phase *morphotopy* and state five conditions that distinguish a morphotopic treatment from a spatial figure of speech, applying them to cases that pass and cases that fail. One structural obstruction, geometric frustration, the failure of locally satisfiable constraints to compose into a global optimum, is definable in any such space; exact enumeration exhibits it in minimal spin models, and its recurrence elsewhere is argued without being demonstrated. Dating sixteen strata shows that 12 fall within the last 197 of 3826 years, with the mean interval between strata contracting from about 1146 years to about 18, an acceleration that partly reflects how finely recent strata are individuated. The contribution is historical and organizational, and a geometric account of the structure of a mind does not explain why there is experience.

## 1. Introduction

Geometry means earth-measure, from *gē* and *metron*, and the etymology records a practice the field left long ago: the survey of land for boundaries, taxation, and the recovery of plots after a flood. The object named has since changed beyond recognition. Working geometers study manifolds no surveyor could walk, including families of probability distributions, latent spaces inside neural networks, and the space of possible animal forms, and none of these is land. The retention of the old name records a function that has been lost, as a vestigial structure does in an organism.

Asking what all these spaces have in common yields an empty answer at that level of generality: they are all called spaces, and the word explains nothing about any of them. The productive question is genealogical. Geometry came to range over morphospaces and embeddings through a sequence of definite changes, each moving its object one step further from the measured field, until the object was a space without a field, then a space without a figure, and finally a space without points. Recovering that sequence shows why the modern uses are continuous with land survey instead of loose analogies to it, and locates the change now under way.

Two commitments organize the account. First, each stratum is treated as a regime determining what could be made visible, and not as a step toward a single correct geometry. Since Euclidean and hyperbolic geometry are each true of their own spaces, the relevant question for each stratum concerns what it allowed a thinker to see that the previous stratum did not, and what it removed

from view in exchange. Second, the sequence is driven by a persistent tension between the figure held fixed and the transformation that deforms it. The tension moves across strata, from a regime in which the figure is primary and transformation threatens its identity to one in which transformation is primary and a stable figure is a slow trajectory. The historical claim is narrow and checkable in its parts: each stratum detaches from one step further from physical extension, and the detachment can be dated (Section 8). The conceptual proposal (Section 7) is that the stratum now forming should be named and equipped with a test that prevents the name from functioning as a slogan.

## 2. Land survey and the Euclidean figure

### 2.1 Geometry as administration

Geometry began as an instrument of administration, and its first objects are those the name still denotes: land, edge, distance, surface. The Greek term for the Egyptian surveyors, *harpedonaptai*, means rope-stretchers, and the reconstruction long associated with them is a rope knotted into twelve equal intervals which, drawn taut into sides of three, four, and five units, forms a right angle. Whether or not the knotted rope was used in this way, the surviving documents establish the character of the knowledge. The Rhind papyrus, copied by the scribe Ahmes around 1650 BCE, takes the area of a circle to be the square of eight-ninths of its diameter, a rule accurate enough for granaries and fields and without any notion of proof. The Moscow papyrus computes the volume of a truncated pyramid, the shape of a real earthwork. In Mesopotamia the Old Babylonian tablet Plimpton 322, from around 1800 BCE, tabulates what amount to Pythagorean triples in base sixty, a millennium before Pythagoras, as a working table and not as a theorem.

In these sources geometry is a technology of place serving the state and astronomy. It fixes boundaries so that possession is stable, computes areas so that grain can be taxed, and orients temples and calendars to the stars. Its propositions are recipes justified by use, and its space is the one the surveyor occupies, defined by the property that it can in principle be crossed and measured. Every later stratum detaches from this intuition of a walkable, measurable space, and its persistence beneath the abstractions explains why mathematicians still draw diagrams of spaces that have none.

### 2.2 Euclid's axiomatic abstraction

The second stratum was founded by removal. Euclid's *Elements*, composed around 300 BCE, does not measure a field (Euclid 1956). It begins from twenty-three definitions, five postulates, and five common notions, and derives some 460 propositions in a deductive order in which each follows by proof from its predecessors and nothing is granted by inspection of a drawn figure. The point that has no part, the breadthless line, and the circle whose radii are all equal are objects available only to reason, and the diagram serves as an aid while the objects themselves are abstract. Form has separated from matter. The triangle of the *Elements* cannot erode, be redrawn larger, or be wrong, because every property of a surveyed plot that could change has been removed from it, and what remains is necessity.

The price of that necessity was contact with the world, which later strata took some 2,000 years to restore, and Euclid himself marked the point where the restoration would begin. Four of his postulates are immediate. The fifth, which in effect states that through a point not on a given line there passes one and only one parallel, is long and conditional, and from antiquity it was felt to be of a different character, more a theorem lacking a proof than an axiom. Proclus recorded this unease in the fifth century, and it persisted. The attempt to derive the postulate from the others eventually overturned the assumption that there is only one geometry.

### 3. Coordinates and non-Euclidean geometry

#### 3.1 The algebraic representation of form

In the two millennia after Euclid little of comparable depth was added, an interval quantified in Section 8, and the third stratum opened with Descartes' *La Géométrie* of 1637 (Descartes 1637). Its decisive step was small and far-reaching. Descartes fixed a unit length, which allowed the product of two segments to be construed as another segment instead of an area, so that the operations of algebra apply directly to geometric magnitudes. A curve became the set of points whose coordinates satisfy an equation, a geometric problem became the manipulation of that equation, and the degree of the equation classified the curve. Fermat reached an equivalent method in the same years. Form became computable, which is the precondition for every later space defined by numbers instead of drawings.

This representation had a consequence that later strata exploited. Once a figure is an equation, its tie to the visible plane is broken: an equation in three unknowns describes a surface, and an equation in more unknowns describes a figure that cannot be seen. The coordinate method thus both captures form and begins to release it from the dimensions of sight, and the higher-dimensional spaces that the nineteenth and twentieth centuries treated as ordinary are licensed in principle once a curve becomes a formula.

#### 3.2 The parallel postulate

The pluralization of space proceeded through a failed proof. In 1733 the Jesuit Girolamo Saccheri, in *Euclides ab omni naevo vindicatus*, set out to vindicate Euclid by assuming the parallel postulate false and deriving a contradiction (Saccheri 1733). In the relevant case he assumed that many parallels pass through the point and derived theorem after theorem, expecting an absurdity that never came. He proved, without recognizing them as such, a long series of true propositions of the geometry he intended to refute, and ended the effort by declaring the conclusion repugnant to the nature of the straight line, an aesthetic judgment in place of the contradiction he never reached.

Others accepted what Saccheri rejected. Lobachevsky published a geometry in which the postulate fails in the *Kazan Messenger* of 1829 (Lobachevsky 1829), Bolyai did so independently in an appendix to his father's textbook in 1832, and Gauss, as his papers later showed, had reached the same results privately and withheld them. For decades the status of these systems remained unsettled, since a body of theorems without evident contradiction is not thereby known to be consistent.

Beltrami settled the question. In 1868 he showed that Lobachevsky's geometry is realized on a surface of constant negative curvature, the pseudosphere, and constructed the disk and half-plane models now bearing his and Poincaré's names (Beltrami 1868), in which the lines and distances of the new geometry are defined in terms of the old. The construction establishes relative consistency: if Euclidean geometry contains no contradiction, neither does hyperbolic geometry, because any contradiction in one would translate into a contradiction in the other. The new geometries thus became as sound as Euclid's, intuition ceased to be the criterion of geometric truth, and the single space became a plurality (Torretti 1978). Historians of mathematics read the episode as part of the onset of mathematical modernism, the turn by which mathematics took its own constructions, and not the physical world, as its subject (Gray 2008).

### 3.3 Riemann's intrinsic geometry

Riemann gave the plurality its general form in a single lecture. His *Habilitationsvortrag* of 1854, printed in 1868, proposed that space is a particular case of a multiply extended magnitude, whose geometry is set by a rule for infinitesimal distance, the line element  $ds^2 = \sum_{i,j} g_{ij} dx^i dx^j$ , in which the coefficients  $g_{ij}$  may vary from point to point. Curvature is then a function of position computed from the  $g_{ij}$  alone. The precedent was Gauss, whose *Theorema Egregium* of 1828 showed that the curvature of a surface is intrinsic, determinable by measurements within the surface and unchanged by any distance-preserving bending, so that an observer confined to the surface could determine it without reference to a surrounding space (Gauss 1828). Riemann removed the surrounding space entirely. A manifold of any dimension has its own metric and curvature, independent of any embedding it may lack, and the metric is, in his words, a matter to be determined within the manifold itself (Riemann 1868; Riemann 1873, trans. Clifford). The fixed ground of Euclid became a varying field, and the geometry the surveyor took as an immovable background became a variable.

### 3.4 General relativity

This abstraction remained internal to mathematics for sixty years. In the general relativity of 1915 and 1916, Einstein identified the gravitational field with the curvature of a four-dimensional space-time manifold, whose metric is determined by the distribution of matter and energy through the field equations, and identified the motion of free bodies with the geodesics of that metric (Einstein 1916). Gravitation, which Newton treated as a force acting across space, became a property of the geometry of spacetime, and the predicted bending of light near the Sun was confirmed in 1919. The geometrization applies to the gravitational field and not to its source: matter and energy enter the equations from outside, as what curves the metric, and are not themselves geometric. With that qualification, the episode is the strongest evidence that the abstractions of this stratum had physical content, since a piece of geometry developed without physical motive turned out to be the language of gravitation.

## 4. Transformation groups and topology

### 4.1 Klein's Erlangen program

By the late nineteenth century there were many geometries, and the question was what unified them. Klein's Erlangen program of 1872 answered by reversing the order of explanation. A geometry, he proposed, is given by a space together with a group of transformations acting on it, and its subject is the properties "not altered by the transformations of the group" (Klein 1872). The group is primary, and a property counts as geometric insofar as the group preserves it. This resolves the tension of the earlier strata into a single structure, because an invariant is defined by the transformations it survives and has no content apart from them; the fixed and the moving become two aspects of one object.

The program orders geometries by inclusion of their groups, and the ordering runs against naive expectation. Projective geometry (Desargues 1639) has the largest group and the fewest invariants, retaining incidence and the cross-ratio while giving up distance, angle, and even the distinction between parallel and intersecting lines. Affine geometry, with a smaller group, recovers parallelism and ratios along a line. Euclidean geometry, whose group of rigid motions is the smallest, has the most invariants, including length, angle, and congruence. A property belongs to a more general geometry the larger the group that preserves it, so the metric notions that seem most basic are the most special. Through Klein's collaboration with Sophus Lie the program also tied geometry to the theory of continuous transformation groups, and symmetry, which later organized physics and machine learning, entered geometry as a defining principle.

### 4.2 Topological invariants

Topology extends the regime to the point where measurement disappears and identity remains. Its oldest result is Euler's, that for a convex polyhedron the numbers of vertices, edges, and faces satisfy  $V - E + F = 2$ , a quantity unaffected by any deformation that preserves the pattern of incidences while distorting every length and angle. For a closed orientable surface the relation generalizes to  $V - E + F = 2 - 2g$ , where the genus  $g$  counts the handles, and the genus alone classifies such surfaces up to continuous deformation: a sphere is not a torus because no deformation without tearing turns zero handles into one, and a coffee cup is a torus because the handle is its only topologically relevant feature. Poincaré's *Analysis Situs* of 1895 founded the systematic study, attaching to a space algebraic invariants, the fundamental group and the homology groups, which record its connectedness and holes and are preserved under continuous maps (Poincaré 1895). Form became identity under deformation, preserved through stretching and broken only by cutting, and the metric that earlier strata had made intrinsic was set aside as inessential to shape.

### 4.3 Symmetry in machine learning architectures

Klein's definition has proved durable well beyond geometry. In the account of Bronstein and colleagues (2021), the dominant architectures of machine learning are instances of the Erlangen prin-

ciple, each determined by the symmetry group of the data it processes. A convolutional network is built around translation, sharing weights across positions so that a pattern recognized in one part of an image is recognized anywhere; graph networks and the set-processing core of a transformer are built around permutation, so that the output does not depend on an arbitrary ordering of the inputs; networks on curved domains are built around rotations of a local frame, the gauge symmetry of the manifold. The choice of transformations under which a network's outputs must be invariant or equivariant is the choice of its inductive bias. The design principle of these machines is thus nearly identical to a definition Klein stated in 1872 for plane figures.

## 5. State spaces

### 5.1 Phase space

The next stratum turns the possible states of a system into a space and its history into a path. In the formulation consolidated by Gibbs in 1902, a mechanical system with  $n$  degrees of freedom is a single point in a phase space of  $2n$  dimensions, one axis for each position and momentum, and its evolution is a curve through that space generated by the Hamiltonian flow (Gibbs 1902). The geometry of the flow carries the physics. Liouville's theorem states that the flow preserves phase-space volume, so that a region of possible states is deformed but never compressed, which is the geometric content of information conservation in reversible dynamics. Poincaré's recurrence theorem, from work of around 1890, states that almost every trajectory on a bounded energy surface returns arbitrarily close to its starting point, a statement about the topology of the flow with no counterpart in the language of forces. The vocabulary of dynamical systems, attractors toward which trajectories are drawn, basins of states sharing an attractor, and bifurcations at which the qualitative portrait of the flow changes, is geometric, and its referents are possible evolutions of a system. Change itself became a geometric object.

### 5.2 Information geometry

The same procedure reaches a domain with no evident spatial character, families of probability distributions. Information geometry, as formulated by Amari and Nagaoka (2000), treats a parametrized family of distributions as a manifold whose points are distributions and whose metric is the Fisher information, which measures how sharply a change of parameter changes the distribution and hence how distinguishable nearby distributions are from samples. The Kullback–Leibler divergence, the natural measure of separation between two distributions, is not a metric, since it is asymmetric and violates the triangle inequality, yet it induces on the manifold a pair of mutually dual flat affine connections, a structure richer than the single Levi-Civita connection of Riemannian geometry. The practical consequence is the natural gradient, a learning rule that follows steepest descent with respect to the Fisher metric instead of the arbitrary coordinates of the parameters, and is therefore invariant to the parametrization of the model. Inference and learning become motion on a manifold of distributions.

### 5.3 Spaces without points

At the far end of this stratum the geometric object loses its points. Grothendieck's reconstruction of algebraic geometry, systematized in the topos theory presented by Mac Lane and Moerdijk (1992), replaces a space considered as a set of locations with the system of its parts and the ways they cover one another. A sheaf on such a system records local data that agree on overlaps and glue together, a topos is a category of such sheaves, and it behaves formally like a space, with its own internal logic, while in general having no underlying set of points. Space becomes a mode of coherence among local pieces. By the end of this stratum the geometric object need not be visible, need not carry a metric, and need not have points; it must only carry structure preserved by some class of transformations.

## 6. Spaces of life and mind

### 6.1 Morphospace

In biology a set of forms constitutes a space. Raup's analysis of shell coiling (1966) parametrizes the logarithmic-spiral shells of molluscs and other groups by four numbers: the whorl expansion rate  $W$ , the distance  $D$  of the generating curve from the coiling axis, the translation rate  $T$  along that axis per revolution, and the shape  $S$  of the generating curve. A choice of the four specifies a shell, and the resulting space, of which Raup plotted slices as a cube, contains every shell the developmental rule can produce. The most informative feature of the construction is its empty regions. Real ammonoids, gastropods, and bivalves cluster in some regions and avoid others entirely, and a form that is geometrically available but never realized indicates a developmental or functional constraint that excludes it. Theoretical morphology, the program McGhee (1999) names and surveys, makes this its method: construct the space of forms a generative rule allows, then ask why organisms occupy the part of it that they do.

The stratum carries its own correction, which matters for the criteria of Section 7. Mitteroecker and Huttegger (2009) show that morphospaces are not in general Euclidean, that distances between forms and the meaning of a straight line between them depend on how the space was constructed, and that the intuition of evenly spaced coordinates misleads where it is most tempting. A morphospace is a space in the full sense only to the extent that its construction licenses the distances and directions read from it; where that license is absent, the spatial picture is decorative. This point becomes the most restrictive of the conditions in Section 7.

### 6.2 Conceptual and semantic spaces

In cognitive science the corresponding territory is meaning. Gärdenfors's conceptual spaces (2000) model a concept as a region in a space whose axes are quality dimensions, in place of a list of necessary and sufficient features, and the substantive claim is that natural concepts occupy convex regions, so that any form lying between two instances of a concept is itself an instance. The clearest case is colour: hue, saturation, and brightness form a space in which the colours grouped under

one word fall in a convex region, with a prototype, the focal example, at its centre and membership grading toward the boundary. Similarity becomes proximity, and a category acquires a shape.

Distributional semantics makes the same structure empirical and scalable, on the principle that a word's meaning is reflected in the contexts in which it occurs. Latent semantic analysis (Landauer and Dumais 1997) extracts a space from word co-occurrence patterns across documents; the neural language model of Bengio and colleagues (2003) learns a distributed vector for each word as a by-product of predicting text; and the word vectors of Mikolov and colleagues (2013) make such embeddings cheap and sharp enough that analogies appear as displacements, the step from man to woman roughly parallel to the step from king to queen. The parallelogram structure is real and repeatable but limited, fragile under changes of corpus and prone to spurious cases. Embeddings show that meaning learned at scale leaves a geometric trace rich enough to compute with. They do not by themselves show that the geometry performs the computation, and that distinction is the subject of the next subsection.

### 6.3 Representational geometry as mechanism

The decisive recent development is that representational geometry can be read from working systems, in brains and machines, and intervened upon, so that it functions as the computation itself. In neuroscience, representational similarity analysis (Kriegeskorte et al. 2008) characterizes a neural population by the matrix of dissimilarities among its responses to a set of stimuli, making the representational geometry of a brain region a measured object comparable across regions, animals, and models. In artificial networks the evidence is stronger, because the system is fully observable and editable. Gurnee and Tegmark (2023) recover from a language model linear directions encoding the latitude and longitude of places and the dates of events, with individual units tracking these coordinates. Li and colleagues (2023) train a network only on sequences of legal moves in the board game Othello, without ever showing it a board, and find in its activations a representation of the current position that a probe can read and that, when edited, changes the model's subsequent play to match the altered board.

In arithmetic the pattern becomes a mechanism. Nanda and colleagues (2023), reverse-engineering a small transformer trained on addition modulo a prime, find that it embeds each number as a point on a circle given by the sine and cosine of an angle and computes sums by composing rotations through the trigonometric identities for the sine and cosine of  $a + b$ , then reading the angle of the result. Engels and colleagues (2024) find the same circular structure arising spontaneously in large models for days of the week and months of the year, irreducibly two-dimensional features used for modular arithmetic over the calendar. Kantamneni and Tegmark (2025) generalize the circle to a helix, showing that a model represents integers along an axis linear in the number and wound with several periods at once, and adds two integers by rotating their helices into the helix of the sum, an operation they call the Clock algorithm and confirm by intervening on the helical directions. Arithmetic is thus carried out as trigonometry, and the claim is causal, since perturbing the geometry changes the answer. Huh and colleagues (2024) argue that the convergence is general: networks trained on different data and even different modalities come to measure similarity among

inputs in increasingly similar ways, as if approaching a shared representational geometry. In the clearest of these cases the geometry is the mechanism that performs the computation, open to measurement and intervention.

#### 6.4 Navigation in morphospace as basal cognition

Levin's framework (2022) gathers the strata of life and mind into a single claim: problem-solving takes place in the anatomical, physiological, and transcriptional spaces of the body as well as in an animal's behavioural space, so that a developing or regenerating tissue that navigates toward a target form and corrects displacements from it exhibits basal cognition, defined as the competent steering of a system through a problem space toward adaptive regions despite perturbation. The planarian flatworm makes the claim concrete. The target morphology toward which its cells rebuild after amputation is encoded in part in a stable pattern of bioelectric voltage across the tissue, in addition to the genome, and a brief manipulation of that pattern can permanently rewrite the target: a cut fragment regenerates two heads in place of one and, left undisturbed, continues to regenerate two heads in later rounds of cutting, with its genome unchanged (Durant et al. 2017). Levin describes the process geometrically: cells minimize the anatomical distance in morphospace between the current and the target form. The setpoint is a point in the space of possible bodies, the bioelectric field acts as a controller, and regeneration is navigation. The organism, the concept, the word, and the developing body thus share a single form, a trajectory in a space of possibilities moving under constraint toward regions that constitute success for that system.

### 7. Morphotopy: definition and criteria

#### 7.1 Definition

The strata of Section 6 no longer share an object with earth-measure. They study how forms become possible, how they persist and deform, and how systems move among them under constraint, and the procedure that produces them has no settled name. We propose *morphotopy*: the study of how forms, organisms, meanings, and agents occupy, generate, and traverse spaces of possible form under constraint. Its object is the production and navigation of spaces. Geometry asks what the structure of a given space is; morphotopy asks how a space of possibility comes to exist for a system, which forms can occupy it, which transformations it admits, which constraints bound it, and which of its regions count as destinations for that system.

#### 7.2 Five conditions

A term of this breadth is justified only if it does more than relabel, and the literature already contains many nominal geometries in which the spatial language does no work and could be deleted without loss. The proposal is therefore stated as a test of five conditions that a treatment must meet to be morphotopic. First, there must be a space of possible forms or states that includes unrealized ones, so that what never occurred still has a location. Second, there must be a notion of nearness or cost, a topology or a metric, whose claims about distance and direction are licensed by the con-

struction and not assumed; this is the condition the morphospace literature isolates (Mitteroecker and Huttegger 2009). Third, there must be a class of admissible transformations, the moves available to the system. Fourth, there must be constraints that bound the space and exclude regions of it. Fifth, there must be a notion of navigation, the goals, attractors, or valued regions toward which trajectories run, so that motion has a relevant direction.

Real cases pass and fail these conditions in instructive ways. Raup's morphospace passes all five: the cube of unrealized shells is the space, the developmental parameters license a meaningful nearness, growth supplies the transformations, the empty regions are the constraints, and selection toward viable forms is the navigation. A bare word embedding passes the first condition and, with care, the second, but fails the fifth until a task is added, since a static cloud of vectors has no destinations; the embedding is an instrument, and the trained network that uses it to compute is the morphotopic object. A loose claim that some domain is a landscape, with peaks and valleys named but no licensed metric and no specified admissible moves, fails the second and third conditions, and the test identifies it as metaphor. The fifth condition separates a morphotopic space from a descriptive one, and the second excludes the largest number of spurious cases.

### 7.3 Relation to existing programs

Morphotopy so defined is broader than several existing programs and contains them as cases. Geometric deep learning applies the Erlangen principle to network architecture, a morphotopy of one engineered domain. Representational geometry measures the spaces inside brains and trained models and is the empirical component. Conceptual spaces are the cognitive instance, and Levin's navigation of problem spaces is the biological one. The proposal does not replace any of these, each of which is sharper in its own domain than a general framework can be. Its contribution is limited to two points: that these programs are instances of one historical movement with a datable trajectory, and that a criterion distinguishes substantive instances from spatial figures of speech. It leaves the mathematics of each domain unchanged.

### 7.4 Geometric frustration as a shared obstruction

If the strata share a historical movement, they may also share structural content, and at least one obstruction is definable wherever a space of forms is subject to competing constraints, although the computation of Section 8 exhibits it in only one stratum. In a frustrated system the geometry of the interactions prevents all local constraints from being satisfied at once. The canonical case is the triangular antiferromagnet: each pair of neighbouring spins has lowest energy when antiparallel, but three spins around a triangle cannot all be mutually antiparallel, so one bond is always unsatisfied. In the spin ice of certain pyrochlore crystals the analogous local rule produces a very large number of degenerate ground states, and the resulting residual disorder gives a measurable zero-point entropy of the kind Pauling (1935) identified in ordinary water ice (Bramwell and Gingras 2001). Stated without the physics, the condition is

$$\min_{x \in X} \sum_i E_i(x) > \sum_i \min_{x \in X} E_i(x),$$

that is, the best attainable global state is strictly worse than the sum of the separately attainable local optima. The right side is always a lower bound on the left, with equality if and only if a single configuration minimizes every local term simultaneously, the unfrustrated case; strict inequality defines frustration. The statement concerns constrained spaces in general and not magnets in particular, which is why the same obstruction appears under other names in optimization, in constraint satisfaction, and in any system whose local preferences cannot be jointly met, and why a space of possible forms generically admits no configuration that is locally optimal everywhere. Section 8 exhibits the inequality exactly in the smallest system that can carry it.

### 7.5 Further cases: music, phylogenetics, and data analysis

The pattern extends beyond physics, biology, and machine learning, and three cases from otherwise unrelated domains meet the test. A chord of  $n$  notes is a point in the quotient of the  $n$ -torus by the symmetric group,  $T^n/S_n$ , an orbifold whose singular boundaries correspond to chords with doubled or evenly spaced notes; Tymoczko (2006) shows that chords near these singularities admit efficient voice leading, short moves to neighbouring chords, so that the harmonic practice composers reached by ear follows the geometry of the space, with consonant triads clustered near its centre. The possible evolutionary trees on a fixed set of species form the space of Billera et al. (2001), assembled from one orthant per tree topology and glued along the lower-dimensional faces where a branch length reaches zero and the topology can change; the space has globally non-positive curvature, which guarantees a unique shortest path between any two trees and hence a well-defined average of a collection of trees, making the comparison and summary of competing phylogenies a geometric operation. Persistent homology (Carlsson 2009) reads the shape of a dataset by growing a family of spaces that connect points within an increasing radius and recording the connected components, loops, and voids that appear and disappear; features persisting over a wide range of radii are the robust shape, and short-lived ones are noise. Each of the three has a space of possibilities, a licensed metric, admissible moves, constraints, and a notion of which regions matter, and each therefore meets the five conditions.

## 8. Computations

Two small computations accompany the argument, each reproduced by the code in `simulation/`, which writes every number below to `simulation/output/results.json` and the two figures to `simulation/output/figures/`. They illustrate two specific claims, the acceleration of the stratigraphy and the existence of frustration, and neither measures history or mind.

### 8.1 Dating the strata

The first computation assigns to each of sixteen strata a representative work and date, from the earth-measure of the second millennium BCE to the present, and computes their order, span, and intervals. The final stratum is the one proposed in Section 7 and is dated to 2026. The dating is conventional, and the earliest stratum is deliberately coarse, since earth-measure has no single date; only the arithmetic on the chosen dates is exact. The number of strata is also a consequential

choice: individuating recent phases finely while treating antiquity in a few broad blocks builds part of the acceleration into the stratification before any date is read, so the contraction reported below reflects the chosen granularity as well as the history. With these qualifications, the strata span 3826 years, and the largest interval between consecutive strata is the 1937 years from Euclid to Descartes, the long interval described in Section 3. The distribution is strongly skewed toward the present. Taking the non-Euclidean geometry of 1829 as the threshold of the modern regime, 12 of the 16 strata fall within the final 197 years, three quarters of the strata within about the last twentieth of the span. Half the strata exist only by 1895, in a history that begins around 1800 BCE. The mean interval between consecutive strata within each era is about 1146 years before the threshold and about 18 years after it, a contraction by a factor of about 64. The detachment of form from extension has accelerated, and the recency of most strata, several of them still contested, is one reason the stratum now forming has no agreed name.

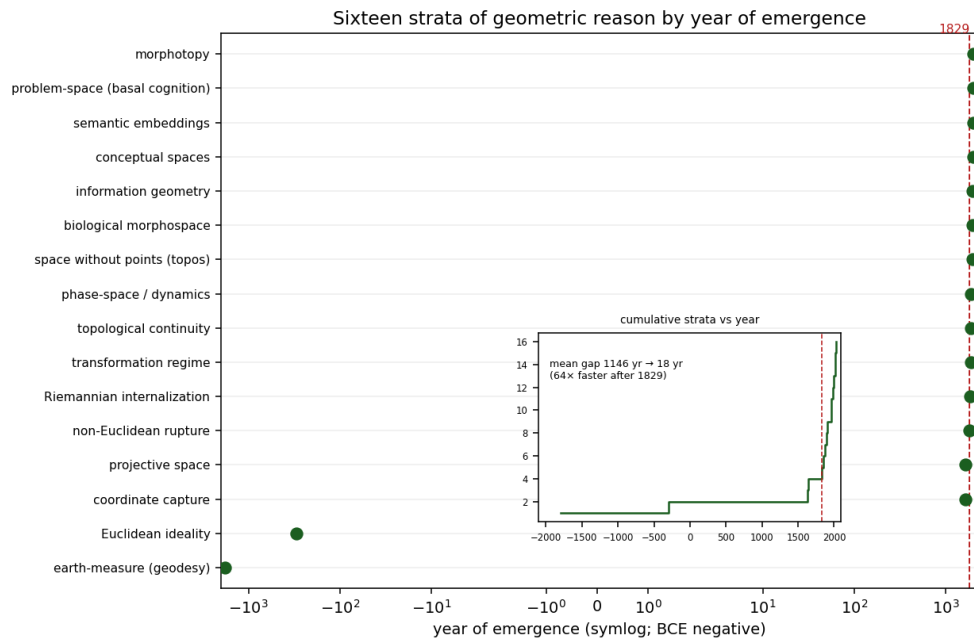


Figure 1: Sixteen strata of geometric reason by date of emergence (symlog axis; BCE negative). Millennia separate the early strata, and twelve of the sixteen fall after the non-Euclidean geometry of 1829. The inset plots the cumulative count against a linear time axis: nearly flat for three millennia and then nearly vertical, with the mean interval between strata falling from about 1,146 years to about 18.

## 8.2 Frustration by exact enumeration

The second computation exhibits the frustration inequality by enumerating all states of small anti-ferromagnetic systems, in which a bond between two spins is satisfied when they are antiparallel. On a triangle, an odd cycle, the three bonds cannot be satisfied together. The lowest attainable energy is  $-1.0$  in units of the coupling, while the three bonds taken separately could reach  $-3.0$ , so the global optimum lies  $2.0$  above the sum of the local optima and the inequality holds strictly.

The ground state is sixfold degenerate, with one of the three bonds unsatisfied in each ground configuration, and the residual entropy is 0.5973 per spin. The square, an even cycle and therefore unfrustrated, is the control: its four bonds are satisfied together, its lowest energy of  $-4.0$  equals the sum of the local minima, no bond is unsatisfied, and its ground state is twofold degenerate, with residual entropy 0.1733 per spin. A single triangle suffices to establish that local optima need not compose into a global one.

Scaling up tests whether this behaviour depends on size. On periodic triangular lattices enumerated exactly at  $N = 9, 12$ , and  $16$  spins, the fraction of bonds unsatisfied in the ground state is one third at every size, the same value as the single triangle. The ground-state degeneracies, 42, 68, and 42, vary non-monotonically with  $N$ , and the residual entropies per spin are 0.4153, 0.3516, and 0.2336, on either side of the exact thermodynamic value of 0.3231 per spin that Wannier (1950) obtained for this lattice. For this small are strongly affected by boundary conditions (the  $4 \times 4$  torus cannot accommodate the three-sublattice ground states), so the enumerations establish the persistence of the one-third frustrated-bond fraction and a finite residual entropy of the right magnitude; they do not show convergence to the thermodynamic limit.

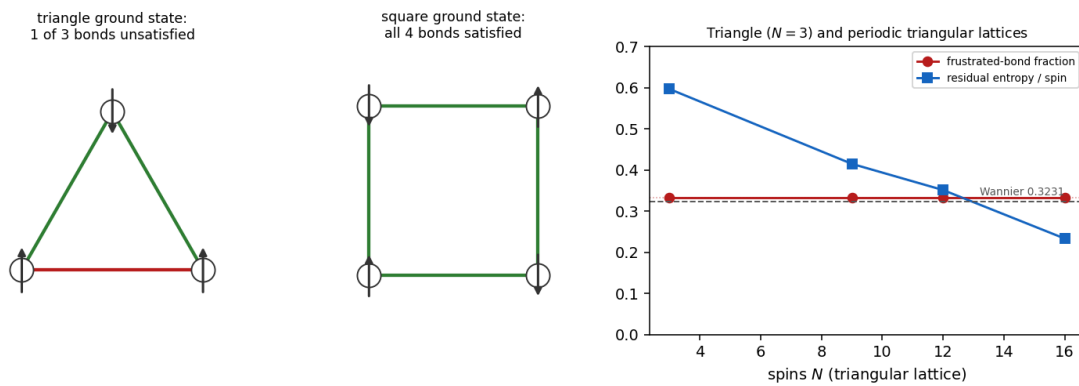


Figure 2: Geometric frustration by exact enumeration. Left: a triangle in a ground state, two spins up and one down, leaves one of its three antiferromagnetic bonds unsatisfied (red), while the bipartite square satisfies all four (green). Right: for the triangle ( $N = 3$ ) and periodic triangular lattices ( $N = 9, 12, 16$ ) the frustrated-bond fraction is one third at every size, and the residual entropy per spin, 0.5973 for the triangle, lies on either side of Wannier's thermodynamic value of 0.3231 on the lattices.

The model does not measure frustration in any applied space. It shows that the obstruction is real and structural, and its occurrence in semantic, developmental, or social spaces must be established case by case.

## 9. Limitations and conclusion

### 9.1 Falsifiability and scope

The proposal is organizational and historical. It is not new mathematics, and it does not predict which domain will next be rendered as a space, although the acceleration in Section 8 suggests that

such renderings will continue. Its empirical content lies in the test, which can fail informatively. If a domain is described in spatial language but no licensed metric and no admissible transformations can be supplied for it, the test excludes the case, and a program of nominal geometries lacking such structure would be identified as decorative. Conversely, the claim that the strata form one movement would be weakened if the cases passing the test shared nothing beyond passing it, and strengthened by a structural obstruction such as frustration that is definable across them (Section 7.4). The criterion is the accountable part of the proposal; the name is open to argument.

## 9.2 Structure and experience

The sequence appears to invite one further claim that should be rejected. A morphotopic account can capture much of the structure of a mind, including the model a system maintains of itself, the integration of its information, the continuity of its trajectory through time, and the value it attaches to regions of its action space, and the work reviewed in Section 6.3 shows how much of that structure is already legible and editable in trained systems. None of it addresses the question Chalmers (1995) isolated, why any such structure is accompanied by experience at all. A geometry of selfhood is not an explanation of consciousness, and the two are easily conflated as the structural account becomes more detailed. Keeping them apart is required by the five conditions, since a morphotopy that claimed to have geometrized experience would violate the demand that its spaces and metrics be licensed by what they explain.

## 9.3 Conclusion

Geometry began with land because land was the first thing that had to be made measurable, divided, owned, and predicted. Its history shows that measurability extends beyond land and beyond physical space: to ideal figures, to curved spaces and spaces without points, to the states of dynamical systems, to the forms a body can take, to the meanings of words, and to the representations that brains and machines build. Morphotopy names the stratum in which the production and navigation of such spaces becomes the object of study, and the five conditions set out here restrict the term to cases in which the spatial structure is licensed by the construction and does explanatory work.

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